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Promoting children's mathematical and statistical understanding through parent-child math games

Mary DePascale [*,1], Geetha B. Ramani

Department of Human Development and Quantitative Methodology, University of Maryland, College Park, USA

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ABSTRACT

Basic statistical literacy is essential for interpreting external sources and developing critical thinking skills necessary for engagement in real-world contexts. However, many children and adults struggle with understanding and interpreting data. Therefore, it is critical to develop engaging, effective methods for teaching early data analysis, as they could enhance children's statistical understanding, math, and higher-order thinking skills. We examined the effectiveness of a home-based, experimental game intervention for children's (ages 5–6, 50 % female, 67 % white, 12 % Asian, 8 % biracial) statistical understanding and math skills. Families (majority high household income and parent education) were randomly assigned to one of three conditions: graphing board game, graphing card game, or literacy board game. Children in the graphing conditions improved on statistical understanding and arithmetic, and children in the literacy condition did not. These results support the development of play-based materials to promote early mathematical and statistical skills, with implications for children's mathematical development.

Basic statistical literacy is essential for understanding and making inferences from information received from external sources and for developing critical thinking skills necessary for engagement in real-world contexts. These skills are essential throughout the lifespan, as adults constantly consider data and sources when making decisions, such as what to buy, where to live, and how to vote. The foundation for these skills is built in early childhood, as data analysis is a central topic area in children's early mathematical development (National Council of Teachers of Mathematics, 2000).

1. Early statistical understanding

In early childhood, data analysis is focused on the abilities to interpret and construct graphs (Franklin et al., 2007; Franklin & Mewborn, 2008; National Council of Teachers of Mathematics, 2000; National Governors Association Center for Best Practices & Council of Chief State School Officers, 2010). As the focus of the current study is on early childhood, we use the term data analysis to refer to these graphing skills. Data analysis is an application of children's foundational math skills, including counting, numerical magnitude comparison, and arithmetic (National Governors Association Center for Best Practices & Council of Chief State School Officers, 2010). For example, a kindergarten class could do an activity where they taste different kinds of apples (e.g., red, green,

* Correspondence to: Department of Human Development and Quantitative Methodology, University of Maryland, College Park, 3942 Campus Drive, College Park, MD 20742, United States.

E-mail addresses: mdep@umd.edu, mdepascale@albany.edu (M. DePascale).

[1] Present address: Department of Educational and Counseling Psychology, University at Albany, State University of New York, Albany, NY, United States.

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yellow), consider which is each person's favorite, and then categorize and use this information to make a picture graph, showing one apple for each student (Whitin & Whitin, 2003). They could also represent this information as a bar graph. In addition to constructing these graphs, their data analysis would also involve interpreting the graphs. This could include answering questions such as: Which color has the most? Which color has the least? Do any colors have the same number? How many green apples are there? Does yellow or green have more? How many more yellow are there than red? How many green and red are there all together? Questions could also relate to constructing graphs, for example: We counted that four students like red apples, does this graph show that? It is clear that engaging in data analysis in this way involves applying foundational math skills, such as magnitude comparison (e.g., which has more yellow or green?) and arithmetic (e.g., how many more yellow are there than red?). Further, understanding and interpreting data contextualizes numbers and mathematical relations and is important across academic subject areas, including math, science, reading, and social studies, as well as real-world experiences (Niezgoda & Moyer-Packenham, 2005; Van de Walle et al., 2018).

The ability to understand and interpret data is also important for productively engaging in these real-world contexts, as it is a critical aspect of thinking critically and drawing inferences and conclusions (Basile, 1999; Larson & Whitin, 2010). Early statistical understanding involves higher-order thinking skills, as children learn how to ask and answer statistical questions and think critically about what is involved in the processes of developing and answering these types of questions (Cook, 2008; Glazer, 2011; Niezgoda & Moyer-Packenham, 2005; Whitin & Whitin, 2003). These abilities to interpret and make inferences from graphs continue to be essential throughout the lifespan (Franklin et al., 2007; Sharna et al., 2010).

Although data analysis is considered a central component of early math learning, it is often not a primary focus of instruction in early grades (English, 2012, 2013), and is often not a focus of research on early math development. Further, despite the lifelong importance of graphing skills and statistical literacy, research shows that middle school, high school, and college students struggle with interpreting and constructing graphs (Lapp & Cyrus, 2000; Leonard & Patterson, 2004; Padilla, 1986; Tairab & Al-Naqbi, 2004). These findings highlight the importance of children having a strong foundation in their early graphing and data analysis skills. Accordingly, additional experience with data analysis and graphing in early childhood is critical, because it has the potential to enhance children's abilities from a younger age (Martinez & LaLonde, 2020).

Specifically, developing engaging, effective methods for teaching early data analysis is important, as they have the potential to enhance children's development of statistical understanding and overall math abilities. In addition, engaging, effective methods can provide children with a stronger foundation for continuing to develop their skills throughout their later schooling experiences and adulthood. In particular, prior research emphasizes the importance of active, meaningful practice with graphing skills (English, 2012; Glazer, 2011; Roth & McGinn, 1997) and considers the role of scaffolded discussions about graphing and data topics for children's learning (Lee & Francis, 2018; Russell, 2006). In considering active and engaging methods for promoting children's graphing and data analysis skills, the current study extends prior work on the role of games as an engaging method for early math learning, by focusing specifically on promoting children's early statistical understanding through a home-based graphing game intervention.

2. Games and playful learning

Games and play are prevalent in early childhood education settings as they provide an engaging context for early learning of social, emotional, and cognitive skills (Golinkoff et al., 2006). In the context of early math learning, play is important as it directly builds on children's math interests and further incorporates math into their daily lives. The use of play and games to promote children's learning builds from classic developmental theories, including sociocultural and cognitive developmental theories, which highlight the importance of developmentally appropriate instruction and adult guidance of children's learning (Vygotsky, 1986) as well as children's engagement in their own learning (Piaget, 1950). More recent playful learning theories describe games and play as an engaging context for learning social, emotional, and cognitive skills (Golinkoff et al., 2006; Hassinger-Das et al., 2017).

Building from the Science of Learning, it is argued that games provide a learning setting where parents or teachers can engage children in child-directed math instruction, as well as a context in which children are active and engaged, as gameplay allows children to directly practice and explore concepts in a structured, dynamic way (Zosh et al., 2018). Math games can provide an opportunity for children to actively practice math and problem-solving skills and a context for parents and teachers to support children's learning by providing examples and feedback through play in an engaging, motivating setting (Clements & Sarama, 2014).

Research also indicates that the home environment is an important context for children's development of early math skills through play and games, and playing games at home supports children's math learning (e.g., Benavides-Varela et al., 2023; Sonnenschein et al., 2016). Studies examining children's home math environment often include games as a part of the landscape of opportunities children may have to learn math at home. As the home math environment relates to children's concurrent and later math outcomes (Daucourt et al., 2021), there is clear potential for math gameplay at home to contribute to children's early math development. In fact, there is correlational evidence that children's engagement in math games at home as preschoolers and kindergartners has been shown to relate to their concurrent math skills and predict their math skills longitudinally through first grade (Niklas & Schneider, 2014; Zhang et al., 2020).

Experimental studies specifically examining the role of games for children's math development have found that playing math games with parents, peers, or teachers can increase children's math knowledge in early childhood, and that even brief sessions (15–20 min) of gameplay can have a profound impact on young children's math skills. Game interventions have shown improvements with medium to large effect sizes in children's counting, numeral identification, magnitude comparison, number line understanding, and arithmetic skills as well as performance on standardized tests of math achievement. These results are consistent across numerical board games (e.g., Skillen et al., 2018) and card games (e.g., Ramani & Scalise, 2020), as well as for children of varying levels of math ability (Siegler & Ramani, 2009), ages (Ramani & Siegler, 2008), and socioeconomic backgrounds (Ramani & Siegler, 2011).

2.1. Math games: Content areas, types, and features

The majority of studies examining math games have focused on games designed to promote children's numerical skills, such as counting (Laski & Siegler, 2014), magnitude comparison (Scalise et al., 2017), number line understanding (Elofsson et al., 2016), and arithmetic (Guberman & Saxe, 2000). While these areas of math are important, early math also comprises topics including algebra, geometry, measurement and data analysis, which along with number and operations, all fall under the umbrella of early problem solving (Ginsburg et al., 2008; National Council of Teachers of Mathematics, 2000). To explore the breadth of the impact of math games, more research is needed on the effect of games for math concepts in these other domains (DePascale & Ramani, 2023). For example, it is possible that games may be effective at promoting skills in each of these math topic areas; however, it is also possible that games may be more suitable for supporting more foundational skills (e.g., counting, numeral identification), skills focused on numerical content (e.g., magnitude comparison, arithmetic), or skills that are more primary focuses of instruction in early grades (e.g., number and operations), as direct practice and use of these skills can be incorporated into game play (e.g., counting each game board space while moving a token, identifying numerals on the spaces). Because of the importance of developing an early understanding of graphing and data analysis (e.g., Glazer, 2011; Martinez & LaLonde, 2020), the current study expands on prior research on math games by examining the use of games for promoting children's statistical understanding.

Additional research is also needed to examine how game designs and features can influence children's learning. Studies have demonstrated that certain elements of games can be designed to facilitate more meaningful practice and interactions that lead to greater growth in children's math skills, including game board designs (Siegler & Ramani, 2009) and game materials and structure (Chao et al., 2000). According to the Cognitive Alignment Framework, games can promote learning through the alignment of the physical representations of concepts and actions of gameplay with children's developing mental representations of the concepts (Laski & Siegler, 2014). Consistent with this theory, prior research indicates that game structures providing more direct practice with relevant math content (e.g., linear versus circular board games for linear number line understanding; Siegler & Ramani, 2009) are known to promote greater learning for the math content area practiced (i.e., linear number games promote greater learning of the magnitude of numbers similar to a number line representation).

However, there is limited work considering how the type of game played impacts children's learning. While some prior studies of math games have included some type of control game or activity, these have either been not math-related (e.g., color; Ramani & Siegler, 2008) or included math content presented in a non-game context (e.g., Siegler & Ramani, 2009). To further understand how games promote children's math learning, studies that compare multiple games designed to teach the same math content area across game types (e.g., board games and card games covering the same math content area) are needed (DePascale & Ramani, 2023). Considering game type is important because game types have different structures, and therefore different affordances, which can directly influence children's learning. Specifically, it is possible that certain topics may align more with a certain type of game design or format based on the types of numerical representations and affordances that each format provides (Laski & Siegler, 2014).

For example, prior studies of math board games identify the affordances of math board games as providing structured, direct practice with number concepts, including numeric magnitudes and the base ten number system. Specifically, horizontal, linear board games provide direct practice with linear number skills and implicitly represent these concepts through the game materials and game play actions (Cheung & McBride, 2017; Ramani & Siegler, 2008; Ramani & Siegler, 2011; Ramani et al., 2012; Siegler & Ramani, 2008; Siegler & Ramani, 2009). In addition, 10×10 grid games implicitly represent and explicitly provide practice with the base ten system through the layout of the game board and the actions of playing the game (Laski & Siegler, 2014; Skillen et al., 2018; Sonnenschein et al., 2016). Within these math board games, features of the board game play (e.g., counting on instead of counting from one, moving one space per unit) direct attention to these key concepts as well (Laski & Siegler, 2014).

For card games, prior studies of math card games identify the affordances of card games as including multiple representations of magnitudes (i.e., symbolic and nonsymbolic representations; Cheung & McBride-Chang, 2015; Ramani & Scalise, 2020; Scalise et al., 2017). In addition, these studies also highlight the role of adult feedback on magnitude comparisons within the card game play itself as important for children's learning. In this way, both the materials of the math card games and the processes of playing the math card games provide representations of and practice with the numeric magnitude concepts.

Building from this prior work on math board games and math card games, the present study compared a graphing board game and a graphing card game. Although both games focused on graphing skills, we expected that the affordances and structure of the graphing board game would promote children's ability to construct graphs and that the affordances and structure of the graphing card game would promote children's ability to interpret graphs. Both of these skills are key components of children's early statistical understanding (Franklin & Mewborn, 2008), and it is important to understand how each type of game may promote these skills differently.

3. Current study

The goals of the current study were to examine (1) the effectiveness of games for promoting children's early understanding of graphing and data analysis in a home-based family math game intervention and (2) how the type of game (e.g., board game, card game) could impact children's learning. Aligned with the Science of Learning and Playful Learning frameworks (Zosh et al., 2018), the games used in the current study were designed to provide children with active, engaging, and meaningful practice with two foundational concepts related to graphs—making graphs and interpreting information presented in graphs (Franklin & Mewborn, 2008)—in a context of play that is socially interactive, joyful, and iterative. The graphs included in the games put numbers into familiar contexts that are relevant across subject areas and life experiences, including graphs about favorites (e.g., foods, books, sports), family size, weather, and going to the zoo, among other topics. Focusing graphing on these familiar contexts allows children to further develop their math skills by making connections between their numerical skills (e.g., counting, magnitude comparison, arithmetic), and applied contexts for problem solving that build on their natural interests (Basile, 1999).

The games were developed to provide specific, direct practice with key elements of graphing (e.g., considering representations of quantities, comparing magnitudes of categories, constructing graphs based on data, applying numerical and mathematical skills, engaging with features of graphs, such as axes). These elements were present through the game design and directly practiced through the actions of playing the game, which involve repeated cues to the concepts, physical representations of concepts, and feedback. The features of games designed to promote learning were specifically designed to be consistent with the affordances and features highlighted in prior work on board games (e.g., implicitly representing features of and providing direct practice with the content area; Laski & Siegler, 2014) and card games (e.g., providing multiple representations of magnitudes; Ramani & Scalise, 2020). We further describe each of these game features below and in Table S1.

The graphing board game was designed such that children were presented with a graphing context and would use their moves throughout the game to construct a graph based on the data presented in the graphing context. This board game was designed to use features that are similar to prior work on board games such as ones that involve using features of the game board and game play to promote understanding of numerical magnitudes and the base ten number system (e.g., Laski & Siegler, 2014; Ramani & Siegler, 2008). Aligned with these number board games, the graphing board game in the current study built from the affordances of math board games to promote early statistical understanding skills. Specifically, on each turn, children had to consider the number of items that were supposed to be represented in each category and consider this in relation to what was currently represented on their graph game board. Accordingly, the game provides direct practice with using given data to construct a graph. This game structure is consistent with prior work on number board games, in which children directly practice linear numerical magnitude skills by moving spaces along a linear or 10 × 10 grid game board. In addition, similar to how features of number board game play (e.g., counting on, moving one space per unit) direct attention to key number concepts (Laski & Siegler, 2014), features of the graphing board game play (e.g., saying out loud what was added to the graph and what it represents, adding one tile per one unit) direct attention to key graphing concepts (e.g., representing quantities, the correspondence of one unit of data to one unit of representation on the graph) as well. Further, similar to how the structure of 10 × 10 grid game boards implicitly highlight the base ten number system (Laski & Siegler, 2014), the structure of the current graphing game board implicitly highlights elements important for constructing graphs, including starting markings for each category at zero on the axis and the necessity to represent each category individually on the graph. These elements (i.e., providing direct practice, highlighting features of graphs) in concert with the practice of filling in the graph based on the provided context were expected to specifically promote children's abilities to make graphs.

In contrast to the graphing board game, in the graphing card game, children were not given the graphing context, but rather had to use the information presented in the graph to interpret the magnitudes represented for each category on the graph. This card game was designed to use features that are similar to prior work on card games such as ones that involve multiple representations of magnitudes to promote numerical magnitude skills (e.g., Cheung & McBride-Chang, 2015; Ramani & Scalise, 2020; Scalise et al., 2017). Aligned with these number card games, the graphing card game in the current study built from the affordances of math card games to promote early statistical understanding, which applies magnitude comparison skills. Specifically, similar to prior card game studies, numerical magnitudes were represented in multiple ways on the graphing game cards. In the current card game, children could see the magnitude of the categories on the graph in three ways, (1) by considering the amount of space each bar or set of objects occupied for each category, (2) by counting the number of discrete objects or grid spaces there were for each category, and (3) by reading the corresponding numeral for each category on the axis of the graph. Consistent with prior number card games, the processes of the game involved receiving parent feedback on magnitude comparisons within the gameplay itself (e.g., Ramani & Scalise, 2020). These elements (i.e., multiple representations of magnitude, parent feedback provided for comparisons), were expected to promote children's ability to interpret information and compare the magnitudes depicted in graphs.

3.1. Aims and hypotheses

3.1.1. Aim 1. Does playing graphing games lead to improvements in children's statistical understanding and math abilities?

According to playful learning theories, as well as empirical work, math games can promote learning in multiple ways (Fisher et al., 2010; Hassinger-Das et al., 2017; Zosh et al., 2018). Specifically, math gameplay can provide direct practice with concepts, involve multiple, redundant cues to the concepts, physical representations of the concepts that align with children's developing mental representations, and feedback to reinforce these concepts. Further, these are provided within a game context that is structured to promote these skills specifically through the design of game materials and the actions involved in playing the game. Through these mechanisms, games for children's math skills, including counting, magnitude comparison, and arithmetic have been shown to be effective (e.g., Guberman & Saxe, 2000; Laski & Siegler, 2014; Scalise et al., 2017). As games have been shown to effectively promote math learning in these areas, we expected that games would also be effective for promoting children's abilities to interpret and construct graphs in the same ways. Further, as graphing involves applications of math skills, including arithmetic and magnitude comparison (National Governors Association Center for Best Practices & Council of Chief State School Officers, 2010; Van de Walle et al., 2018), we expected that playing graphing games would also promote children's abilities in these areas, as the graphing included in the games inherently allows for applications of arithmetic and magnitude comparison skills.

3.1.2. Aim 2. Does playing a graphing board game versus a graphing card game lead to differences in improvements in children's statistical understanding?

Drawing from the Cognitive Alignment Framework (Laski & Siegler, 2014) and previous empirical work (e.g., Ramani & Siegler, 2011; Siegler & Ramani, 2009), we expected that gains in statistical understanding would differ based on the type of game played due to features of the games themselves and the corresponding practice they provide. As described above, each graphing game's design included and afforded specific features, consistent with prior work on math board games and card games. In line with prior work, these features were expected to promote children's statistical understanding abilities, with features supporting the abilities to interpret (card game) and construct graphs (board game). As these features were specific to the affordances and structure of each game type, we expected that the different game types could promote learning differently (i.e., that card games would promote the ability to interpret graphs and board games would promote the ability to construct graphs).

3.1.3. Exploratory analyses

In addition to analyses for the two primary aims of the study, we also conducted additional, exploratory analyses to further understand children's learning from the games and experiences playing the games. Specifically, we examined relations between dosage of gameplay and children's learning, as previous studies have demonstrated that there is variability in whether dosage impacts children's learning from math games and what outcomes relate to dosage (e.g., Ramani & Scalise, 2020). We also conducted descriptive and exploratory analyses to provide additional information about the context of families' gameplay during the intervention, including parent perceptions of their and their child's experiences with the games and concepts included in the intervention (as in Purpura et al., 2021; Vandermaas-Peeler et al., 2012). As these analyses were exploratory, we did not have specific hypotheses.

Table 1
Descriptive Statistics of Demographic Survey Variables.

	$n = 148$
Child Race, *n* (%)	
African American or Black	5 (3.4)
White	99 (66.9)
Asian American or Pacific Islander	18 (12.2)
American Indian or Alaska Native	1 (0.7)
Biracial/Mixed Race	12 (8.1)
Other	11 (7.6)
Missing	2 (1.4)
Child Ethnicity, *n* (%)	
Hispanic or Latino	16 (10.8)
Not Hispanic or Latino	131 (88.5)
Missing	1 (0.7)
Parent Race, *n* (%)	
African American or Black	6 (4.1)
White	106 (71.6)
Asian American or Pacific Islander	26 (17.6)
American Indian or Alaska Native	1 (0.7)
Biracial/Mixed Race	8 (5.4)
Other	1 (0.7)
Parent Ethnicity, *n* (%)	
Hispanic or Latino	9 (6.1)
Not Hispanic or Latino	139 (93.9)
Parent Education, *n* (%)	
High school diploma/GED	1 (0.7)
Some college coursework/vocational training	6 (4.1)
2-year college degree (Associates)	4 (2.7)
4-year college degree (BA/BS)	47 (31.8)
Postgraduate or professional degree (MA, PhD, MD, JD)	89 (60.1)
Missing	1 (0.7)
Annual Household Income, *n* (%)	
Less than $15,000	2 (1.4)
$15,000 to $30,000	3 (2)
$31,000 to $45,000	5 (3.4)
$46,000 to $59,000	5 (3.4)
$60,000 to $75,000	15 (10.1)
$76,000 to $100,000	18 (12.2)
$101,000 to $150,000	41 (27.7)
$151,000 or more	57 (38.5)
Missing	2 (1.4)
School Experience, *n* (%)	
PreK	31 (20.9)
Kindergarten	67 (45.3)
First Grade	16 (10.8)
Kindergarten or First Grade and Other	4 (2.7)
Other[a]	21 (14.2)
Missing	9 (6.1)

[a] "Other" school experiences included: preschool, outschool classes, homeschool, homeschool due to the pandemic, pod school, and summer break/summer camp

4. Method

4.1. Participants

Data were collected from 148 children ages 5;0 to 6; 11 (M_{age} = 5;11, 50 % female) and their parent. The age range was selected based on the Common Core Standards for kindergarten, first, and second grades for measurement and data, which relate to representing and interpreting data, and include constructing picture graphs and bar graphs and interpreting these graphs using arithmetic and magnitude comparison skills (National Governors Association Center for Best Practices & Council of Chief State School Officers, 2010). An additional 30 children completed a pretest but their data were excluded from final analyses for not completing a posttest. The sample size was determined to be adequate based on guidelines for sample sizes in structural equation modeling (Bentler & Chou, 1987; Kenny, 2015) and prior developmental research with similar sample sizes and methodology (e.g., Ramani et al., 2019).

Participants were recruited from posts on social media, email listservs, and ChildrenHelpingScience.com (Sheskin et al., 2020), as well as information shared by local schools, children's museums, and libraries. In order to be able to mail study materials to participants, participation was limited to participants living in the United States (n = 144) and Canada (n = 4). Parents provided informed consent and children provided verbal assent prior to participation. At the time of consent, parents completed a demographic survey (Table 1).

4.2. Design

The study used an experimental intervention design with three conditions and pretest and posttest assessments of children's knowledge (see Table 2). Children were randomly assigned to one of three game conditions—graphing card game, graphing board game, and literacy board game (active control)—with assignment stratified by children's age and gender to maintain similar numbers of girls and boys and 5- and 6-year-olds in each condition. Children were randomly assigned to condition prior to being administered the pretest.

4.3. Procedure

Families completed a pretest, 4-week intervention, and posttest (data collected 03/2021–09/2021). All test sessions were conducted online via Zoom. At each test session, children completed measures of their statistical understanding and general math ability. During the intervention, parents and children were asked to play a brief (~15 min) game together in their home at least three times per week. All game materials were mailed to families. Families were provided with two versions of the game they were assigned and were asked to play each version for two weeks. Families were given written instructions for playing the games and a video of the experimenter describing how to play the game and demonstrating how to use the materials.

Families were also provided with paper and online versions of a gameplay log to record the days and times they played the game with their child (see Fig. S1). Parents were emailed weekly reminders during the intervention period to remind them to play the game with their child and to fill out their log each time they played.

4.4. Experimental conditions

Participants were randomly assigned to one of three experimental conditions. (1) Graphing Card Game, (2) Graphing Board Game, and (3) Literacy Board Game (Active Control). The graphing card game was focused on interpreting the information presented in graphs, and the graphing board game was focused on constructing graphs from data. As described above, each game type (i.e., board game, card game) included and afforded different features that align with early graphing skills (i.e., constructing graphs, interpreting graphs; see also Table S1). Both games types included picture graphs and bar graphs, because children's early experiences with data build on their experiences with classification of objects, and both picture graphs and bar graphs are a way to represent categories of data. In the current study, picture graph games were played before bar graph games, because bar graphs provide a more abstract representation of the content in picture graphs. Therefore, starting with picture graphs allows children to progress from more concrete to more abstract representations of data (Friel et al., 2001, as cited in Van de Walle et al., 2018). All graphs included in all games represented five categories and had a numerical axis ranging from zero to ten. The average total number of data points represented on the graphs (i.e., sum of all categories) was 26. Table S2 provides a summary of the materials families in each condition used each week

Table 2
Summary of Study Design.

Session	1		N/A	2	
Description	Pretest		Intervention (4 weeks)	Immediate Posttest	
Measures /Activities	Child	Parent	Child and Parent	Child	Parent
	• Statistical understanding • Math ability	• Demographic questionnaire	Play game together at least 3 times per week	• Statistical understanding • Math ability	• Parent perception of gameplay intervention survey

of the intervention.

4.4.1. Graphing card game

For the graphing card games (called *Dare to Compare: Pictures!* and *Dare to Compare: Colors!*), parents and children were given a deck of 25 cards with each card depicting a graph or chart (see Fig. 1a). For the first two weeks of the intervention, the game included cards

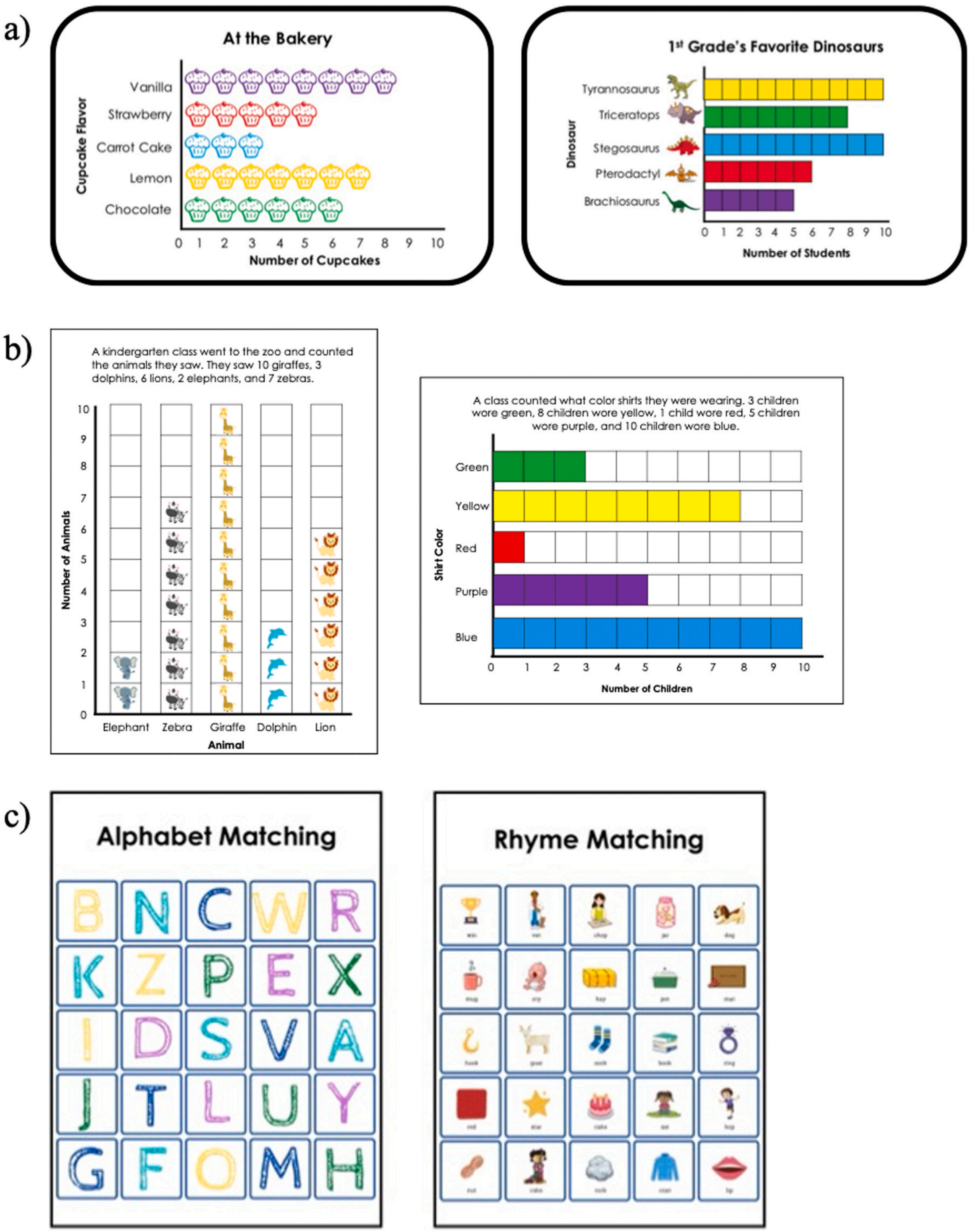

Fig. 1. Intervention Game Materials: (a) Graphing Card Game cards, (b) Graphing Board Games, (c) Literacy Games.

that depicted picture graphs, and for the second two weeks, the game included cards that depicted bar graphs. In both games, each graph represented quantities of five categories of information, and each category was represented with a color (i.e., red, yellow, green, blue, or purple).

To play the game, parents and children would start with the deck face down between them. Players would then each choose a color that would be used throughout the game. On each turn, one player would flip over one card from the top of the deck, put it in the middle, and read the title of the graph out loud. Then, each player would say the number represented on the graph for their color and the category it represented (e.g., for green, "There are 6 chocolate cupcakes at the bakery."). Parents were instructed to prompt their child to count the pictures/color squares if their child was not sure how many their color had. Next, parents would ask their child which category had more (e.g., for green and blue, "Were there more chocolate or carrot cake cupcakes at the bakery?") The player whose color had the larger amount would win the card. If both players' colors had the same amount, they would each flip over an additional card and compare the amounts on that card to determine who would win both cards. Players would continue until the entire deck of cards had been used. To end the game, each player would count the number of cards in their pile, and the player with the most cards would win the game.

4.4.2. Graphing board game

The graphing board games (called *Top the Chart!* and *Raise the Bar!*) involved parents and children rolling dice and collecting tiles to make a graph. Both players each had their own small identical game board. The game boards showed a description of the graph context (e.g., "A kindergarten class went to the zoo and counted the animals they saw. They saw 10 giraffes, 3 dolphins, 6 lions, 2 elephants, and 7 zebras") as well as an empty graph template (see Fig. 1b). Tiles fit into the squares on the graph template and depicted the categories represented on the graph (e.g., giraffes, dolphins, lions, elephants, and zebras). Parallel to the graphing card game condition, for the first two weeks of the intervention period, the game board was for a picture graph, and for the second two weeks, the game board was for a bar graph.

To play the game, parents and children would read the description of the graph context at the top of the board out loud. On their turn, players would roll two dice to determine the actions to complete during that turn. One die indicated the number of pieces to add to their game board (values included 1, 2, and 3). The other die indicated the type of pieces to add to their game board (e.g., the items corresponding to each of the categories represented by the graph). When adding tiles to the graph, each player would describe what they were adding (e.g., "I added 2 giraffe tiles, which shows that they saw 2 giraffes at the zoo."). If the total number of tiles would exceed the number listed in the graph description, a player would add the number needed and discard any remaining tiles (i.e., put them back in the central pile of tiles). If a player already had the number needed for a certain category and they rolled that category (e.g., rolling an elephant when they already had the necessary two elephants), they would skip their turn.

Parents were instructed to prompt their child to count the tiles on the graph if their child was not sure how many tiles they had for a category. Players would continue taking turns until one player successfully completed their graph. To end the game and determine if the graph was completed correctly, each player would count the number of tiles that they had in each category and compare the number given in the graph description (e.g., "My graph shows 10 giraffes, and they saw 10 giraffes at the zoo."). The first player to complete their graph correctly would win the game.

4.4.3. Literacy board game

The literacy board games (called *Match It: Alphabet Matching!* and *Match It: Rhyme Matching!*) were structured similarly to the graphing board games, in that parents and children each had their own game board and were playing to fill in their boards. To complement the other conditions, for the first two weeks of the intervention, the game board was for alphabet matching, and for the second two weeks, the game board was for matching rhyming words (see Fig. 1c).

To play the game, parents and children would start by placing the picture/rhyme cards between each of their game boards. On their turn, players would roll a number die (values included, 1, 2, and 3) to determine how many cards to draw. Players would then draw the number of cards and say what was on their card. For the alphabet matching game, players would label the picture on their card (e.g., "Watermelon") and the letter it started with (e.g., "W"). Then, players would add the card to the matching space on their game board (e.g., the "Watermelon" card would be added to the "W" space). For the rhyme matching game, players would read the word on their card (e.g., "Net") out loud. Then, players would add the card to the matching space on their game boards (e.g., the "Net" card would match the "Vet" space). If a player already had a card covering the matching space for a certain letter/rhyme, they would discard any extra letter/rhyme cards (i.e., add them to a separate discard pile). Players would continue taking turns rolling the dice and adding cards to their boards until one player covered all of the spaces on their board. The first player to cover all of the spaces on their board would win the game. These games were intended to serve as an active control in comparison to the graphing board and card game conditions. In playing, children had a parallel experience of parent-child game play over the four-week intervention, however, the play was not focused on any mathematical or statistical content.

4.5. Materials and measures

At pretest and posttest, children completed measures of Statistical Understanding, Arithmetic, and Magnitude Comparison. The same measures were used at pretest and posttest. For all measures, questions were presented on PowerPoint slides via the share screen function on Zoom, and the experimenter read each question aloud and recorded the child's responses. Pretests were conducted by one experimenter (aware of the hypotheses of the study) and posttests were conducted by five experimenters (52 % of posttest sessions conducted by an experimenter unaware of the participant's condition and the hypotheses of the study).

4.5.1. *Statistical understanding*

Children completed a measure of statistical understanding comprised of 18 items. These items were developed for the current study based on teacher-created activities and materials related to graphing instruction for preschool and elementary school students, sample items from standardized math tests, age-appropriate questions and curricula on instructional websites (e.g., Khan Academy), and online graphing activities for children.

The questions focused on two areas of children's understanding—interpreting graphs (9 items) and constructing graphs (9 items). For each area, questions covered the content for both picture graphs and bar graphs. For interpretation questions, children were shown a graph and asked to answer questions about the information presented in the graph, including identifying which category had the most, comparing magnitudes of categories, and counting or adding total amounts. For questions about constructing graphs, children were given information about a graphing context, shown three graphs, and asked to identify which of the presented graphs was correct for the given context (see Fig. 2). The number of correct responses both overall ($\alpha = .89$) and for each subsection (i.e., interpreting ($\alpha = .79$) and constructing ($\alpha = .82$) graphs) were used as outcome variables.

4.5.2. *Math ability*

Measures of magnitude comparison and arithmetic were used to measure broader math ability. These content areas were selected because graphing abilities build on children's understanding of them (National Governors Association Center for Best Practices & Council of Chief State School Officers, 2010). For example, to interpret information presented in a graph, children need to be able to identify and compare the magnitudes represented for each category. Similarly, arithmetic is involved in understanding the magnitudes of categories represented on graphs, as interpreting graphs includes adding total amounts across categories and adding and subtracting to determine how many more or fewer items one category may have in comparison to another.

4.5.2.1. Magnitude comparison. Children were shown pairs (20 items) or sets of three numbers (16 items) and asked to choose which number was more (e.g., "Which is more, 5 or 6?"; Fig. 3). Sets of numbers (e.g., 10, 15, 8) were used in addition to pairs of numbers, as interpreting graphs can involve comparing the values of more than two numbers or categories. The task included comparisons of the numbers 1 to 9 as well as comparisons of larger numbers, with values ranging from 10 to 81 (adapted from Ramani et al., 2019). For number pairs, ratios ranged from 0.20 (e.g., 1 and 5) to 0.93 (e.g., 82 and 88). Pairs were balanced such that the larger number was presented on the left in half of the pairs. Sets were balanced such that the largest number was presented in the left, center, and right positions each one-third of the time. The total number of correct responses was used as the outcome variable.

4.5.2.2. Arithmetic. Children were shown addition (8 items), subtraction (8 items), and word problems (6 items) one at a time and asked to provide an answer (Fig. 3). Children were asked to complete the problems mentally without writing anything down. Addition problems included problems with two single-digit addends (e.g., $3 + 9$) as well as problems with one single-digit addend and one double-digit addend (e.g., $11 + 5$). Subtraction problems included problems with a single-digit minuend and a single-digit subtrahend (e.g., $8 – 4$), as well as problems with a double-digit minuend and single-digit subtrahend (e.g., $15 – 6$). Word problems included three addition problems and three subtraction problems. The specific problems used were adapted from arithmetic measures for children of similar ages (Elofsson et al., 2016; Ramani et al., 2019) and from problems used in standardized measures such as the Test of Early Mathematics Ability (Ginsburg & Baroody, 2003) and the Woodcock-Johnson IV Calculation subtest (Schrank et al., 2014). The total number of problems correct was used as the outcome variable.

4.5.3. *Dosage*

Dosage of gameplay was calculated from the gameplay logs families filled out. Measures included the number of minutes played and the number of days the game was played.

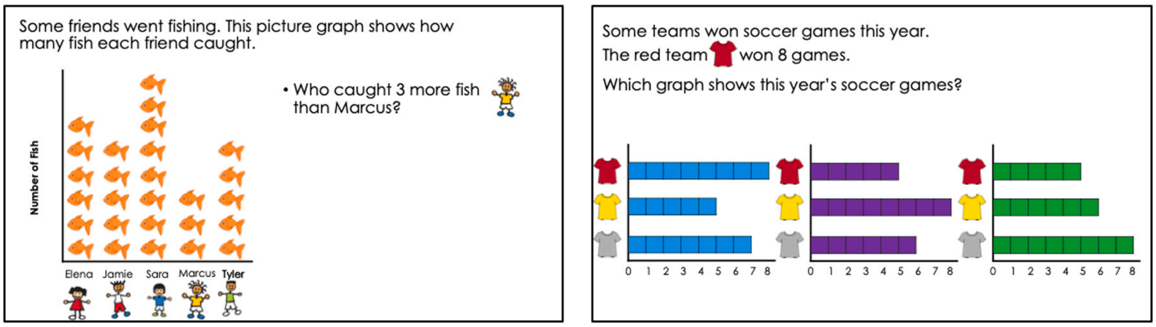

Fig. 2. *Sample Items from the Statistical Understanding Measure. Note.* A sample item for interpreting graphs is shown on the left, and a sample item for constructing graphs is shown on the right. Right: Axis labels from top to bottom show pictures of a Team Red shirt, a Team Yellow shirt, and a Team Gray shirt.

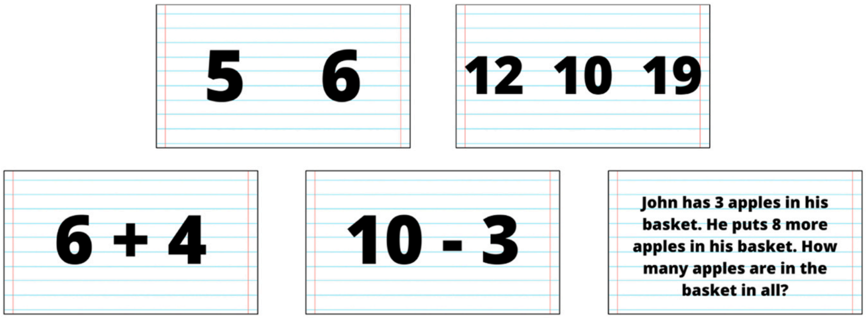

Fig. 3. *Sample Items from the Measures of Math Ability. Note:* Top: Sample magnitude comparison items, including a pair (left) and set of three numbers (right). Bottom: Sample arithmetic items, including addition (left), subtraction (center), and word (right) problems.

4.5.4. *Parent perceptions of gameplay intervention survey*

Parents also completed a survey after their child completed the posttest, to provide additional information about the context of gameplay. Specifically, the survey included items for parents to rate their and their child's enjoyment of the games, and indicate their perception of children's familiarity with the game concepts (e.g., picture graphs, bar graphs) prior to playing the games as well as their perception of any changes in their child's understanding of these concepts after playing the games. The same items were used for all conditions. Items and descriptive statistics for each condition are provided in Tables 3 and 4.

4.6. *Data analysis*

The first aim was to examine if playing graphing games led to improvements in children's ability to construct and interpret graphs. Structural equation modeling (SEM) was used to examine Aim 1. SEM analyses were conducted using Mplus Version 8.7. SEM involves two modeling phases—a measurement model which shows how measured variables are indicators of latent variables, and a structural model which shows relations among latent variables. In the current study, multigroup second-order latent growth models were used to examine changes in children's statistical understanding, arithmetic, and magnitude comparison over the two study time points (Curran & Hancock, 2021; Ram & Grimm, 2007; Serang et al., 2019, i.e., from pretest to posttest. Separate models were used for each measure (i.e., statistical understanding, arithmetic, magnitude comparison).

Prior to fitting the growth models, confirmatory factor analysis models were fit to examine the fit of the measurement models to the data. Separate models were used for each measure. Fig. S2 shows the confirmatory factor analysis models for statistical understanding, arithmetic, and magnitude comparison. Measured variables are shown in rectangles and latent variables are shown in circles. Factors included ability at pretest (time 1) and posttest (time 2). For statistical understanding, these were indicated by interpreting graphs and constructing graphs variables. For arithmetic, these were indicated by addition, subtraction, and word problems variables. For magnitude comparison, these were indicated by single-digit pairs, double-digit pairs, single-digit sets, and double-digit sets variables. For each factor, corresponding factor loadings were constrained to be equal, to assume invariance across time points as the same measures were used at each time point (Hancock et al., 2001). In addition, corresponding residuals (e.g., pretest interpreting graphs and posttest interpreting graphs) were set to covary, as the same participants completed measures at each time point (Hancock et al., 2001).

Table 5 shows a summary of fit indices for the models. The indices used to evaluate data-model fit included the root mean squared

Table 3
Parent Perception of Gameplay Intervention Survey Items (Enjoyment and Familiarity).

		Board Games			Card Games			Literacy Games		
Item	Description	*n*	*M*	*SD*	*n*	*M*	*SD*	*n*	*M*	*SD*
Enjoyment[a]	Please rate how much you and your child enjoyed the games.	-	-	-	-	-	-	-	-	-
Item 1	How much did you enjoy the games?	47	3.68	0.86	46	3.54	0.86	48	3.50	0.85
Item 2	How much did your child enjoy the games?	46	3.85	1.05	46	3.50	0.81	48	3.67	1.16
Familiarity[b]	Please indicate how familiar your child was with the following concepts prior to playing the games.	-	-	-	-	-	-	-	-	-
Item 1	Picture graphs	47	2.94	1.11	46	2.67	1.23	48	3.50	0.85
Item 2	Bar graphs	47	2.79	1.10	46	2.65	1.30	48	3.67	1.16
Item 3	Reading the axes of a graph or chart	47	2.36	1.05	46	2.33	1.30	48	3.27	1.23
Item 4	Comparing numerical magnitudes	47	3.74	1.19	46	3.50	1.28	48	2.94	1.16
Item 5	Letters and letter sounds	47	4.51	0.80	46	4.54	0.72	48	2.40	1.09
Item 6	Rhymes and rhyming words	47	4.43	0.83	46	4.43	0.66	48	3.60	1.27

[a] Enjoyment items were rated on the following scale: (1) Really did not enjoy the games, (2) Did not enjoy the games, (3) The games were okay, (4) Enjoyed the games, (5) Really enjoyed the games.
[b] Familiarity items were rated on the following scale: (1) Very unfamiliar, (2) Unfamiliar, (3) Somewhat familiar, (4) Familiar, (5) Very familiar.

Table 4
Parent Perception of Gameplay Intervention Survey Items (Change in Understanding).

	Board Games				Card Games				Literacy Games			
Item, Rating[a]	-1	0	1	Missing	-1	0	1	Missing	-1	0	1	Missing
Picture graphs, *n* (%)	0 (0)	11 (23.4)	35 (74.5)	1 (2.1)	1 (2)	7 (13.7)	37 (72.5)	6 (11.8)	0 (0)	33 (66)	6 (12)	11 (22)
Bar graphs, *n* (%)	0 (0)	6 (12.8)	40 (85.1)	1 (2.1)	0 (0)	7 (13.7)	39 (76.5)	5 (9.8)	0 (0)	34 (68)	5 (10)	11 (22)
Reading the axes of a graph or chart, *n* (%)	1 (2.1)	9 (19.1)	36 (76.6)	1 (2.1)	0 (0)	9 (17.6)	34 (66.7)	8 (15.7)	0 (0)	32 (64)	8 (16)	10 (20)
Comparing numerical magnitudes, *n* (%)	0 (0)	27 (57.4)	19 (40.4)	1 (2.1)	1 (2)	23 (45.1)	21 (41.2)	6 (11.8)	0 (0)	35 (70)	5 (10)	10 (20)
Letters and letter sounds, *n* (%)	0 (0)	38 (80.9)	6 (12.8)	3 (6.4)	0 (0)	37 (72.5)	5 (9.8)	9 (17.6)	0 (0)	32 (64)	14 (28)	4 (8)
Rhymes and rhyming words, *n* (%)	0 (0)	40 (85.1)	3 (6.4)	4 (8.5)	0 (0)	38 (74.5)	5 (9.8)	8 (15.7)	0 (0)	19 (38)	27 (54)	4 (8)

[a] All items were in response to the question: “Please indicate if you noticed changes in your child’s understanding of the following concepts over the 4 weeks” and were rated on the following scale: (−1) My child understood less after playing the games, (0) My child’s understanding did not change after playing the games, (1) My child understood more after playing the games, (NA) Not applicable

error of approximation (RMSEA; parsimonious index) and the standardized root mean squared residual (SRMR; absolute index). Guidelines suggest that SRMR values less than or equal to 0.08 and RMSEA values less than or equal to 0.06 are indicative of adequate data-model fit (Hu & Bentler, 1999). Model fit was good for the statistical understanding model, adequate for the arithmetic model, and inadequate for the magnitude comparison model. Because the model for magnitude comparison had poor data-model fit, magnitude comparison was not examined in the growth model phase.

Fig. 4 shows the growth models conducted for statistical understanding and arithmetic. Second-order factors included intercept and growth factors. First-order factors included ability at pretest (time 1) and posttest (time 2). Overall, these models allow for variability in each group to differ, and they allow for examining differences across conditions in any parameter of interest in the model. In the current models, the parameter of interest was the mean growth over time (growth factor). To examine differences by condition in this parameter, each difference (e.g., $M_{Growth:BoardGames} - M_{Growth:CardGames}$) was coded as an additional parameter and tested in the model.

The second aim was to examine if playing a board game versus a card game led to differences in improvements in children's statistical understanding abilities. To examine this question, t-tests were used to compare pretest-posttest gains in interpreting graphs and constructing graphs scores between the board games and card games conditions.

5. Results

5.1. Descriptive statistics

5.1.1. Pretest and posttest measures

Table 6 shows descriptive statistics for pretest and posttest measures of statistical understanding, arithmetic, and magnitude comparison, by condition. Overall, we observed variability in children's abilities; however, for the magnitude comparison measure, the majority of children were at ceiling at pretest. For statistical understanding abilities specifically, we observed variability in children's abilities at pretest, with opportunity for improvement overall, and in both interpreting and constructing graphs.

5.1.2. Dosage of gameplay

In total, 142 families (96 %) submitted logs of their gameplay. Overall, total number of dates played ranged from 0 to 16 ($M = 9.28$, $SD = 3.87$), and total number of minutes played ranged from 0 to 710 ($M = 172.29$, $SD = 102.53$). Table 7 shows dosage by condition.

5.1.3. Parent perception of gameplay intervention survey measures

In total, 141 parents (95 %) completed the parent survey after their child completed the posttest. There was variability in parents' responses. Table 3 shows descriptive statistics for ratings of enjoyment and perception of children's familiarity with the game concepts, and Table 4 shows descriptive statistics for perception of children's change in understanding of the game concepts.

5.2. Preliminary analyses

Preliminary analyses were conducted to examine if there were any differences across conditions in children's statistical understanding at pretest. However, we did not expect pretest differences, because children were randomly assigned to conditions. One-way ANOVAs indicated that there were no differences by condition at pretest in children's overall statistical understanding scores ($F(2145)= .36$, $p = .698$), interpreting graphs scores ($F(2145)= .33$, $p = .721$), or constructing graphs scores ($F(2145)= .39$, $p = .679$). Therefore, pretest ability was not included as a covariate in subsequent analyses.

In addition, one-way ANOVAs were used to examine if there were differences by condition in annual household income and dosage of gameplay, to determine if these variables should be included as covariates in subsequent analyses. Results indicated there were no differences by condition in income ($F(2143)= .89$, $p = .412$), total dates played ($F(2139)= .70$, $p = .500$), or total minutes played ($F(2139)= .36$, $p = .696$). Therefore, annual household income and dosage of gameplay were not included as covariates in subsequent analyses.

Finally, prior to conducting the analyses for Aim 1, the Mahalanobis distance was calculated for each participant from their pretest scores on statistical understanding, arithmetic, and magnitude comparison to determine if there were any multivariate outliers (consistent with best practices for structural equation modeling; Mueller & Hancock, 2018). Based on the distances calculated, one participant was determined to be an outlier (Distance=31.31, $p < .001$) and was therefore not included in analyses for Aim 1.

Table 5
Summary of Data-Model Fit for Measurement and Growth Models.

Model	df	χ^2	RMSEA [90 % CI]	SRMR
Measurement Models				
Statistical Understanding	7	2.78	0.000 [0.000, 0.073]	0.033
Arithmetic	33	42.31	0.076 [0.000, 0.137]	0.053
Magnitude Comparison	75	224.28	0.202 [0.171, 0.232]	0.274
Growth Models				
Statistical Understanding	7	2.79	0.000 [0.000, 0.073]	0.033
Arithmetic	33	42.31	0.076 [0.000, 0.137]	0.053

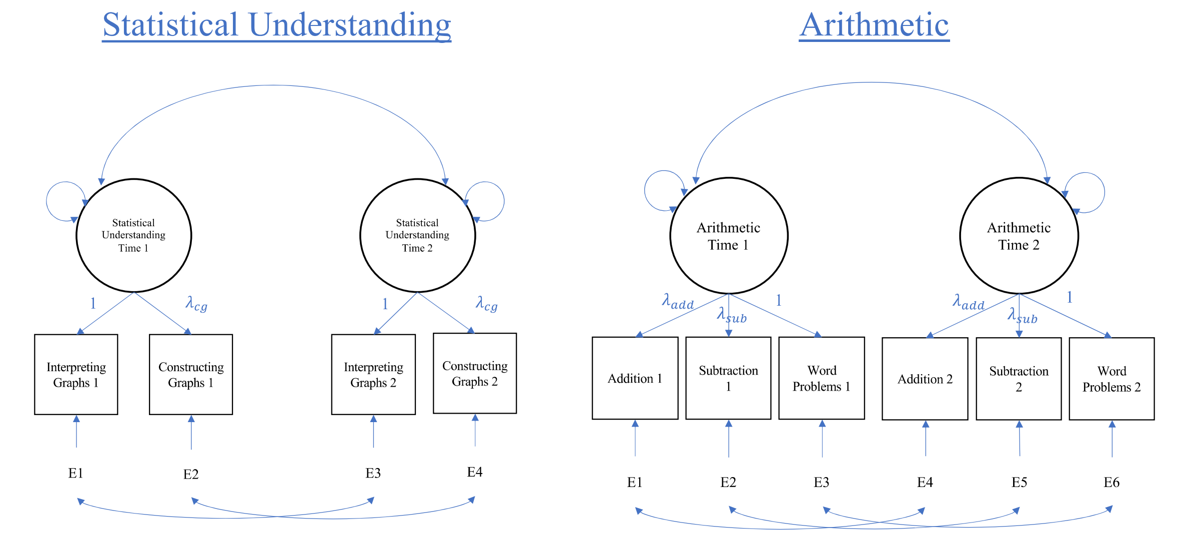

Fig. 4. *Multigroup growth models used to examine Aim 1. Note.* Models were identical for each condition.

Table 6
Descriptive Statistics of Pretest and Posttest Measures, by Condition.

***Time*, Measure**	**Board Games**		**Card Games**		**Literacy Games**	
	M	***SD***	***M***	***SD***	***M***	***SD***
Pretest						
Statistical Understanding (Overall)	61.94 %	25.61 %	61.67 %	27.00 %	65.78 %	28.33 %
Statistical Understanding (Interpreting Graphs)	64.11 %	25.89 %	62.11 %	28.44 %	66.44 %	26.56 %
Statistical Understanding (Constructing Graphs)	59.78 %	29.11 %	61.22 %	29.89 %	65.11 %	33.22 %
Arithmetic	47.00 %	33.77 %	43.59 %	32.68 %	54.91 %	31.55 %
Magnitude Comparison	89.89 %	14.19 %	83.06 %	19.92 %	89.06 %	15.19 %
Posttest						
Statistical Understanding (Overall)	71.61 %	23.44 %	71.00 %	26.28 %	70.22 %	27.72 %
Statistical Understanding (Interpreting Graphs)	71.11 %	24.11 %	72.33 %	24.56 %	71.11 %	27.44 %
Statistical Understanding (Constructing Graphs)	72.11 %	25.89 %	69.67 %	86.00 %	69.33 %	33.22 %
Arithmetic	50.41 %	33.41 %	47.23 %	33.68 %	56.64 %	31.45 %
Magnitude Comparison	90.19 %	13.97 %	84.83 %	16.22 %	89.39 %	16.75 %

Note: We report the percent correct for each measure in this Table to simplify interpretation. However, the raw data was used in all analyses.

Table 7
Dosage of Gameplay by Condition.

Variable, Condition	*n*	min	max	*M*	*SD*
Total Dates Played					
Board Games	45	1	13	9.60	3.34
Card Games	48	0	16	9.52	4.24
Literacy Games	49	0	16	8.76	3.95
Total Minutes Played					
Board Games	45	0	267	163.55	70.62
Card Games	48	0	710	171.06	110.88
Literacy Games	49	0	585	181.54	118.79

5.3. Primary analyses

5.3.1. Aim 1

The first aim was to examine if playing graphing games led to improvements in children's statistical understanding and math abilities. Multigroup second-order latent growth models were used to examine differences in statistical understanding and arithmetic between each of the study conditions. Indices for each model are shown in Table 5. Model fit was good for the statistical understanding model and adequate for the arithmetic model.

Table 8 shows a summary of the results for each model. For both the statistical understanding and arithmetic models, results indicated that the mean of the growth factor was significant for both the board games ($M_{Statistical} = 0.77, p < .001, r = 0.50; M_{Arithmetic} =$

Table 8
Summary of Growth Model Results.

	Board Games			Card Games			Literacy Games			$D_{Board–Card}$			$D_{Board–Literacy}$			$D_{Card–Literacy}$		
	Est.	SE	*p*	Est.	SE	*p*	Est.	SE	*p*	Est.	SE	*p*	Est.	SE	*p*	Est.	SE	*p*
Statistical Understanding																		
$M_{Intercept}$	5.72	0.32	< .001 * *	5.77	0.31	< .001 * *	6.00	0.33	< .001 * *	-0.05	0.42	.906	-0.28	0.45	.534	-0.23	0.44	.606
$Var_{Intercept}$	3.71	1.32	.005 * *	3.62	1.24	.003 * *	4.62	1.58	.003 * *	0.09	0.93	.925	-0.91	1.08	.399	-1.00	1.10	.364
M_{Growth}	0.77	0.20	< .001 * *	0.73	0.24	.002 * *	0.37	0.19	.050	0.04	0.28	.875	0.40	0.26	.124	0.35	0.26	.172
Var_{Growth}	0.49	0.40	.212	0.39	0.32	.222	0.55	0.74	.460	0.11	0.47	.818	-0.06	0.82	.945	-0.16	0.79	.835
Arithmetic																		
$M_{Intercept}$	2.44	0.28	< .001 * *	2.28	0.26	< .001 * *	2.75	0.25	< .001 * *	0.16	0.38	.672	-0.31	0.37	.403	-0.47	0.36	.187
$Var_{Intercept}$	3.16	0.38	< .001 * *	2.98	0.32	< .001 * *	2.82	0.42	< .001 * *	0.18	0.48	.710	0.35	0.57	.540	0.17	0.54	.756
M_{Growth}	0.19	0.09	.026 *	0.20	0.09	0.034 *	0.10	0.08	.211	-0.003	0.13	.978	0.09	0.12	.451	0.09	0.12	.448
Var_{Growth}	-0.03	0.11	.786	0.15	0.11	.171	0.05	0.09	.559	-0.18	0.15	.242	-0.08	0.14	.559	0.10	0.14	.500

* $p < .05$, * * $p < .01$

0.19, $p = .026$, $r = 0.31$) and card games conditions ($M_{Statistical} = 0.73$, $p = .002$, $r = 0.39$; $M_{Arithmetic} = 0.20$, $p = .034$, $r = 0.29$) but not the literacy games condition ($M_{Statistical} = 0.37$, $p = .050$, $r = 0.27$; $M_{Arithmetic} = .10$, $p = .211$, $r = 0.18$). None of the parameters estimated for the differences between conditions were significant (all M's between -0.003 and 0.34, $p > .05$). Overall, this pattern of results indicates that children in the board game and card game conditions showed significant growth in statistical understanding and arithmetic, and children in the literacy games condition did not; however, the differences in growth were not substantial enough to be significantly different across conditions.

5.3.2. Aim 2

Aim 2 was to examine if there were differences in gains in statistical understanding based on the type of graphing game played (i.e., board game or card game). T-tests were used to compare pretest-posttest gains in interpreting graphs and constructing graphs in the board and card game conditions. Although the average gains in each condition followed the hypothesized pattern (i.e., Interpreting Graphs: $M_{\mathrm{BoardGames}} = 0.64 < M_{\mathrm{CardGames}} = 0.92$; Constructing Graphs: $M_{\mathrm{BoardGames}} = 1.11 > M_{\mathrm{CardGames}} = 0.76$), results indicated that there were no significant differences between the graphing game conditions in pretest-posttest gains in interpreting graphs ($t(96) = -.79$, $p = .429$, $d = -.16$) or constructing graphs ($t(96) = .93$, $p = .355$, $d = .19$).

5.4. Additional analyses

Additional exploratory analyses were conducted to further examine children's learning from the games. These included analyses examining (1) dosage of gameplay and (2) relations with parent perceptions of gameplay and children's learning. Results for each of these analyses are reported in the following sections.

5.4.1. Dosage of gameplay

To further examine relations of children's gameplay and learning, dosage of gameplay (e.g., total dates played and total minutes played) was examined in relation to children's pretest-posttest gain scores. Table S3 shows correlations between these variables, overall and by condition. Overall, dosage did not relate to pretest-posttest gains.

5.4.2. Parent survey measures

Relations between variables of interest and parent perception of gameplay intervention survey measures were also examined to further understand patterns in children's learning from the games. Table S4 shows Spearman correlations between these, pretest-posttest gains, and dosage of gameplay.

Overall, parent reports of their own and their child's enjoyment of the games were significantly positively related ($r(139) = .60$, $p < .01$). Parent and child enjoyment were also significantly related to perceived change in their child's understanding of game concepts (i.e., picture graphs, bar graphs, reading the axes of a graph or chart, comparing numerical magnitudes, letters and letter sounds, rhymes and rhyming words) (r's between .19 and .45, $p < .05$). Child enjoyment was also significantly related to measures of dosage of gameplay ($r(133) = .21$ and .27, $p < .01$). For items about graphing skills, parent report of their children's familiarity with the concepts was significantly negatively related to their perceptions of changes in their child's understanding of the concepts (e.g., parent perception of low familiarity before completing the study related to perception of higher changes in understanding after completing the study) (r's between $-.36$ and $-.25$, $p < .01$). Familiarity with picture graphs and bar graphs also significantly negatively related to dosage of gameplay (r's between $-.26$ and $-.20$, $p < .05$).

For pretest-posttest gains, parent perceptions of change in understanding of bar graphs significantly positively related to gains on all measures of statistical understanding (overall, interpreting graphs, constructing graphs) (r's between .19 and .25, $p < .05$). Parent perceptions of change in understanding of numerical magnitudes significantly positively related to gains in statistical understanding overall scores ($r(129) = .18$, $p < .05$).

6. Discussion

The goals of the current study were to examine (1) if playing graphing games led to improvements in children's statistical understanding and math abilities, and (2) if playing a board game versus a card game led to differences in improvements in children's abilities to interpret and construct graphs. To address these questions, families (majority high household income and parent education) played graphing or literacy games together at home for four weeks. The results of this study provide initial evidence that playing graphing games at home can improve children's statistical understanding abilities.

6.1. Role of graphing games for children's statistical understanding and math abilities

6.1.1. Statistical understanding

Overall, the findings provide preliminary evidence that playing graphing games may lead to improvements in children's statistical understanding abilities. Specifically, findings indicated that there was significant growth in statistical understanding with medium to large effect sizes for children in the board games and card games conditions, but not the literacy games condition, although growth did not differ across conditions. These results indicate that children who played the graphing games showed significant improvement in their graphing skills after completing the intervention. Results also align with theories of playful learning (Fisher et al., 2010; Zosh et al., 2018) and are consistent with previous research indicating that game interventions have led to improvements with medium to large effect sizes in math outcomes (Cheung & McBride, 2017; Laski & Siegler, 2014; Skillen et al., 2018) and that playing numerical games in the home environment can support children's math learning (Benavides-Varela et al., 2016, Cheung & McBride, 2017; Ramani & Scalise, 2020; Sonnenschein et al., 2016). The findings of the current study build on findings from these previous numerical game studies by showing that the home environment is a supportive context for children's learning of early graphing skills through play.

Further, findings contribute to the literature on children's early math development by considering their statistical understanding abilities. Few prior studies have examined this area of young children's math development, even though it is a central component of foundational math learning (National Governors Association Center for Best Practices & Council of Chief State School Officers, 2010). The current study provides both descriptive information about young children's abilities to construct and interpret graphs and information about the role of games in promoting these skills in early childhood. Specifically, we observed variability in children's statistical understanding abilities, including variability and room for improvement in both interpreting and constructing graphs. Across the measures of statistical understanding, at least one third (and up to 40 %) of children answered fewer than 50 % of the questions correctly. Because statistical understanding remains an essential skill throughout the lifespan (Franklin et al., 2007; Sharna et al., 2010), it is critical to understand how to support children's learning in this area.

6.1.2. *Math abilities*

Findings related to gains in children's math abilities were mixed. For arithmetic, we found that there was significant growth with medium effect sizes for children in the board games and card games conditions, but not the literacy games condition. These findings indicate that children who played the graphing games showed significant improvement in their arithmetic skills after completing the intervention, although gains were not substantial enough to differ across conditions. These results add to the literature on math games supporting children's learning of math skills. Previous research has shown that playing math games can promote children's arithmetic (Chao et al., 2000; Cheung & McBride, 2017; Elofsson et al., 2016; Guberman & Saxe, 2000; Ramani & Siegler, 2011); however, the games used in those studies targeted arithmetic skills more directly. In the current study, games targeted graphing skills, however, as graphing skills build on children's arithmetic skills (National Governors Association Center for Best Practices & Council of Chief State School Officers, 2010), additional practice with graphing during gameplay can provide additional practice with arithmetic as well.

It is important to note that while arithmetic could be incorporated into gameplay in multiple ways, arithmetic was not directly required to successfully play the games. For example, in the graphing board game, players could use arithmetic to determine how many total tiles they had in a category after adding tiles to their board on their turn (e.g., I had 3 yellow tiles and I added 2 more yellow tiles, and 3 plus 2 is five, so I have five yellow tiles now) as well as how many more tiles they needed to fill a certain category (e.g., I have 3 blue tiles and I need 10 blue tiles, 10 minus 3 is 7, so I need 7 more blue tiles). However, using arithmetic in these ways was not necessary to play the game. For the card game, players could use arithmetic to consider how many more or fewer players had in each of their categories (e.g., my category has 5 votes and your category has 8 votes, 8 minus 5 is 3, so your category has 3 more votes than mine), but this was not necessary to play the game. In this way, gains in arithmetic after playing the games further demonstrate the connections between children's early graphing and children's early math skills.

For magnitude comparison, gains in children's abilities were difficult to evaluate, as about 51 % of participants answered 35 or 36 out of 36 items correctly at pretest. Model fit for the confirmatory factor analysis model was not adequate, so a growth model was not examined. Future studies could further examine the role of graphing games for children's magnitude comparison abilities in other ways, such as non-symbolic magnitude comparison or considering measures of magnitude comparison beyond accuracy, such as reaction time.

6.1.3. *Game type*

We also examined graphing game type (e.g., board game, card game) in relation to children's learning. Prior research indicates that the structure of a game can influence children's math learning, and it is possible that certain topics may align more with a certain type of game design or format based on the types of numerical representations each format provides (Laski & Siegler, 2014). Based on this prior research on the affordances of math games, the graphing board game focused on constructing graphs and the graphing card game focused on interpreting graphs. Contrary to our predictions, we found that there were no significant differences in gains in children's abilities to interpret or construct graphs between children who played graphing board games and children who played graphing card games. This suggests that playing graphing games can improve children's graphing skills more generally. In addition, it may be possible that practice with either skill (i.e., interpreting or constructing graphs) during gameplay may support children's learning of the other skill. In this way, when children play a game focused on constructing graphs, they could improve their abilities to construct graphs and be able to apply what they learn to their abilities to interpret graphs. Similarly, when children play a game focused on interpreting graphs, they could improve their abilities to interpret graphs and be able to apply what they learn to their abilities to construct graphs. For example, when playing the board game, children could potentially build skills to interpret graphs by checking the value of what is currently represented on their or their parent's graph on a given turn. Similarly, when playing the card game, children could potentially build skills to construct graphs if they or their parent notice and point out features of graphs (e.g., starting each category from zero on the axis) while considering the magnitudes for each of their categories.

6.2. *Dosage of gameplay*

The current study included gameplay logs as a measure of dosage, as understanding the amount of gameplay children participate in is important for understanding patterns in children's learning from the games. Compared to previous studies (Ramani & Scalise, 2020;

Sonnenschein et al., 2016), the percent of gameplay logs returned (96 %) was high. It is possible that offering families the option of completing the log on paper or online contributed to the higher percentage of returned logs.

Overall, there were wide ranges in the number of dates and number of minutes that families played the games. This replicates findings from previous home-based game intervention studies showing variability in dosage (Ramani & Scalise, 2020; Sonnenschein et al., 2016).

In the current study, dosage of gameplay was not correlated with pretest-posttest gains in children's statistical understanding or math abilities. Few previous studies have reported relations of dosage and learning gains. In one study, Ramani & Scalise (2020) found that the number of minutes (but not the number of times) families played a magnitude comparison game related to gains in children's magnitude comparison and number line skills, but there were no relations with gains in other math knowledge measures. They also found that for families who played a shape and color game, number of minutes played related to gains in children's shape finding, and number of times played related to gains in children's counting; there were no relations with gains in other math knowledge measures (Ramani & Scalise, 2020).

In the current study, it is possible that other factors beyond the amount of play may have been important for children's learning from the games. For example, it is possible that interactions during play, the extent to which families played the games according to the game instructions, or other individual differences may have influenced children's learning from the games in addition to the number of times or minutes they played. In addition, because early statistical understanding is often not a focus of instruction in early grades (English, 2012), it is also possible that any amount of practice with the graphing concepts from the games in the current study could support children's learning and development of statistical understanding skills.

Exploratory analyses also indicated that dosage was positively related to parents' ratings of children's enjoyment of the games and was negatively related to parents' perception of children's familiarity with picture graphs and bar graphs before the intervention. These findings suggest the possibility that some of the variability in dosage may be due to parents adapting the amount of gameplay based on their perceptions of their children's playing and learning experiences.

6.3. *Context and experiences of gameplay: Parent survey measures*

After participants completed the posttest, parent survey measures were collected in order to provide further information about families' experiences playing the games and parents' perceptions of children's skills before and after playing the games. Specifically, we considered three sets of questions which measured parent and child enjoyment of the games, parents' perceptions of children's familiarity with the game concepts prior to completing the study, and parents' perceptions of changes in children's understanding of the game concepts over the four weeks of playing the games. Consistent with the use of similar measures in prior studies (e.g., Purpura et al., 2021; Vandermaas-Peeler et al., 2012), these survey measures can serve as an indicator of the context of families' gameplay and participation and experiences with the intervention. In this way, parent survey measures provide a better understanding of the intervention in the home environment and can inform future math game interventions.

6.3.1. *Parent and child enjoyment*

Overall, the majority of parents reported that they (55 %) and their child (59 %) enjoyed or really enjoyed the games. There were no significant differences in enjoyment across conditions, indicating that each of the games was considered enjoyable for parents and children. This is also important given the relations of enjoyment and dosage described above, as differences in enjoyment could impact how often families choose to play the games. In addition, for the graphing games, there were no differences in enjoyment of the games based on children's initial statistical understanding ability at pretest. This is important, as it suggests that each game was engaging regardless of initial ability. In contrast, if enjoyment had differed based on ability level, this could indicate that the content of the game was too easy or difficult based on children's initial understanding, which could make the games less engaging for children.

Results also indicated that parent and child enjoyment were significantly related to parents' perceptions of changes in children's understanding of the game concepts (e.g., picture graphs, reading axes of a graph or chart, comparing numerical magnitudes, letters/letter sounds, rhymes/rhyming words). This finding aligns with theories of playful learning, as engagement and joy are central components of how playful experiences promote learning (Zosh et al., 2018).

6.3.2. *Perceptions of familiarity and change in understanding of game concepts*

Parents were asked to report their perceptions of how familiar their child was with the game concepts prior to playing the games and if they noticed changes in their child's understanding of the same concepts over the four weeks of playing the games. Note that in the current study, these measures were completed at the end of the study. We included the familiarity measure at the end of the study because we did not want parents' reflections on their child's familiarity with the concepts to influence their engagement with the games. Although we did not observe statistically significant differences between conditions in parents' report of their child's familiarity with the game concepts, it remains possible that their participation in the intervention could have retrospectively influenced their perception of their child's familiarity prior to playing the games.

For perceived familiarity, parents reported that children were relatively unfamiliar with graphing concepts before playing the games. Specifically, the percent of parents reporting that children were familiar/very familiar with each concept was the following: picture graphs (32 %), bar graphs (28 %), and reading the axes of a graph or chart (14 %). In contrast, the majority of parents reported that children were familiar/very familiar with comparing numerical magnitudes (56 %), letters and letter sounds (87 %), and rhymes and rhyming words (86 %). Lower levels of familiarity with graphing concepts are consistent with previous research indicating that data analysis and statistical literacy are not often a primary focus in early math instruction (English, 2012, 2013). Further, lower familiarity with these concepts highlights the need for additional methods, such as graphing games, for teaching graphing concepts in early childhood.

For perceptions of change in children's understanding, the majority (range: 76 % to 87 %) of parents in the board games and card games conditions reported that children understood each of the graphing concepts more after playing the games. This is important, as it indicates that parents noticed changes in their children's abilities and suggests that they viewed the games as effective for promoting learning of the concepts. Notably, parents' perceptions of children's change in understanding of bar graphs significantly related to children's actual pretest-posttest gains in statistical understanding (both overall and for interpreting graphs and constructing graphs scores). This finding suggests that parents' perceptions of changes in broader graphing concepts (e.g., picture graphs, bar graphs) related to the overall concepts measured in the statistical understanding task (i.e., interpreting graphs, constructing graphs).

6.4. Limitations

The current study has several limitations. There was approximately 17 % attrition (30 participants completed a pretest and received game materials, but did not complete a posttest). While this amount is low in comparison to other home-based game intervention studies (Scalise et al., 2022; Sonnenschein et al., 2016), it is still a sizeable amount. While many factors impacting attrition were outside of control of the study (e.g., participant health and family emergencies, impacts of the COVID-19 pandemic), other aspects of the study could be changed to aim to reduce participant attrition. For example, studies could reduce the suggested amount of gameplay (e.g., play 2 times/week instead of 3 times/week). Similarly, instructions to families could further emphasize that they should feel free to play as little or as much as is reasonable for their family, which may make the study more feasible and help with participant retention.

In addition, although steps were taken to reach and recruit a diverse sample of participants, the final sample was diverse geographically, but did not otherwise represent a diverse sample. The majority of participants were white, not Hispanic/Latino, had high household income, and were highly educated (91 % had a 4-year college degree or a postgraduate degree), which can limit generalizability of the findings. Future studies should aim to reach a more diverse sample of participants. This is especially important for understanding how children learn from the games and developing game materials that are enjoyable and accessible for children. For example, prior research with math games has shown that, when matched on initial knowledge level, children from low-income backgrounds benefited more from playing math games than children from middle-income backgrounds (Ramani & Siegler, 2011). One reason for this could be variability in prior experience with games and gameplay (Ramani & Siegler, 2011). In the current study, it is also possible that different subgroups of children may learn differently from the games, based on varying experience with games or gameplay at home. Understanding these differences is important for developing games to best support children's learning.

6.5. Future directions

The current study opens several areas for future research. As described above, although we observed significant gains in statistical understanding and arithmetic skills within the graphing games conditions, gains were not substantial enough to differ across conditions. One possibility for this pattern of results is that the current study focused on the home environment, which leads to numerous potential contributors to variability (e.g., how the game was played, distractions while playing the game). Future studies could examine gameplay in different settings, which may have less variability in the gameplay context. Specifically, examining children's learning from play with an experimenter would allow for examining children's learning when other aspects of gameplay (e.g., dosage, guidance and feedback during play, adherence to game instructions) are kept constant. To further understand the effects of play across conditions, future studies could also examine these effects in multiple additional ways. For example, although the sample size in the current study was adequate for the analyses conducted, it is still relatively small for structural equation modeling. Future studies could consider effects with a larger sample size. In addition, future studies could consider effects over a longer period of time in a delayed posttest, as prior studies have observed growth in children's abilities after the intervention period (e.g., Muñez et al., 2022).

In addition, future work could consider the role of individual factors in relation to families' engagement in gameplay and children's learning from the games. Parent factors (e.g., math anxiety, attitudes and beliefs about math or learning through play) have the potential to influence the amount of time parents engage in math gameplay with their children and the types of interactions they have with their children during gameplay. For example, parents may engage differently in the games based on whether they believe play should be more parent-led or child-led, as well as whether or not they believe play is an important context for children's learning.

Further, studies could also consider parents' math and stats abilities as well. Given research suggesting that high school and college students struggle with interpreting and constructing graphs (Lapp & Cyrus, 2000; Leonard & Patterson, 2004; Padilla, 1986; Tairab & Al-Naqbi, 2004), it is possible that parents may also struggle in these areas. Importantly, the games used in the current study focus on graphing content that is simple for adults (e.g., placing the same number of tiles that is written in the data into an empty graph template, comparing relative magnitudes of different categories). However, if parents vary in their understanding or confidence in their own abilities for these concepts, this could influence gameplay in multiple ways. First, parent ability could influence dosage of play or affect during play if parents feel they do not understand enough to play the games. Second, previous research has shown that parent math ability relates to both parent and child use of advanced math talk during play (DePascale et al., 2021). In this way, it is also possible that parent abilities could influence the content of parent-child talk during play, which could then influence what and how children learn from the games.

Future studies could also further examine children's learning from the games. For example, studies could examine the role of children's initial statistical understanding level in their learning from the games, as learning may differ based on initial knowledge level (Ramani & Siegler, 2011; Siegler & Ramani, 2009). In addition, studies could include an additional time point (e.g., delayed posttest) to examine if changes in children's abilities endure over time. Studies could also consider how children's learning from games compares to their learning from other materials or instructional activities (e.g., a worksheet). Understanding if and how children's learning differs between different playful (e.g., games) and direct instruction (e.g., worksheet) contexts would allow for a deeper understanding of the role of games in children's development of statistical understanding and math abilities.

Future studies could also further examine children's statistical understanding abilities. As described above, we observed variability in both children's abilities and parent-reported familiarity with graphing concepts. Future work could examine factors that contribute to this variability, such as school experience, the amount and type of graphing- and data-related content children have engaged with (e.g., curricula used, informal learning experiences), child math ability and executive function skills, and parent ability, attitudes, and beliefs related to these concepts and the importance of these concepts for children's learning. In addition, to further understand children's abilities in this area, studies could also examine children's understanding of different types of graphs (e.g., picture graph, bar graph) and factors that influence their abilities to interpret and construct these types of graphs. For example, children's attention to and memory for features that are figurative versus abstract (e.g., color) varies (see Chan & Mazzocco, 2017; Lange-Küttner et al., 2023); and this could impact the way that they engage with and understand picture graphs and bar graphs, as these graph types vary in how abstractly they represent information. In line with this, future work could also consider if interventions (e.g., games) targeting only picture graphs, only bar graphs, or both picture graphs and bar graphs are most effective at promoting children's early statistical understanding skills.

6.6. Conclusion

Basic statistical literacy is essential for understanding and making inferences from information received from external sources and for developing critical thinking skills necessary for engagement in real-world contexts throughout the lifespan. Although math games are known to be an effective method for instruction in early childhood, the current study is the first to examine the role of games for mathematical topics beyond early numerical skills, specifically statistical understanding. Because graphing and early data analysis skills are essential for children's math development and later statistical literacy, the use of games to further engage children and promote their learning of these skills is important. The findings provide initial evidence that children may learn from playing graphing games with parents at home, as well as descriptive information about children's early statistical understanding skills and parent and child engagement in graphing games at home. These results and future replications of them may have implications and applications for families and educators, as the types of games used in the current study could be incorporated into classroom or home activities to engage children in their learning of these concepts in a playful, enjoyable way. Results can also support the development of future play-based interventions and materials to promote children's early mathematical and statistical skills, with implications for children's later development and achievement.

CRediT authorship contribution statement

Geetha B. Ramani: Writing – review & editing, Methodology, Conceptualization. **Mary DePascale:** Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Conceptualization.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.cogdev.2024.101480.

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